

UNIT-IV

Advanced concepts:

Terminology & concepts:

- The BC economy is triggering the invention of many new ideas & the reappropriation of existing concepts & terminology in innovative ways.
- It prompts investigating the definition of terms that have been taken for granted & passed unquestioned for years, such as money, currency, property, gov't, sovereignty, & intellectual property.
- The questioning of underlying definitions & the reappropriation of terms position these concepts more openly & accessibly for applⁿ to current situations.
- BC-related concepts are more actively in people's minds & ready to apply at generalized level.
- Conceptualization is an intellectual process that refers to careful analysis of general ideas i.e. concepts to more clear, more distinct implementations.
- BCT, generalized conceptualization process involves careful analysis of 2 fundamental aspects
 1. BC
 2. Bitcoin or plainly coin.

→ BCT:

- BC is a technology for decentralized ledger.
- Decentralization could be instantiated in many ways, such as smart property, delegati democracy, governance services & community-based credit bureaus.

- New models like BC force us to consider reality at more generalized level of the concepts behind a specific instantiation Eg: Library system
- At more generalized conceptual level, a library is a system of value exchange, thus one product & service offerings like books & research journals.
- This leads us to imagine other specific situations that could be realized with BC concepts.

Bitcoin or coin:

- Bitcoin is instantiation of decentralization as a digital currency.
- It can be instantiated in ~~as~~ many ways, like smart property, delegate democracy, governance services & community-based credit bureau.
- An individual can earn some coin by performing service like mining or crowdfunding that person can burn or spend.
- In concept generation, one can immediately understand the innovations of coin ideas.
- precedentcoin is aptoken or ~~remuneration~~ remunerative coin for performing function of establishing precedence.
- Thus in generalized conceptualization process, there is a need to have tools for realizing possibility.
- New conceptualization can shift thinking back & forth b/w levels of general & specific.

Economy as an ex:

- A specific vs general thinking is notion of an economy

- An economy at immediate level is people buying & selling things
- An economy at higher, more generalized conceptual level is production & consumption of things.

→ Blockchain as an Ed:

- BCR at immediate, specified level is a decentralized public ledger for recording of cryptocurrency transaction.
- BCR at higher level is new class of things like
 - The Internet
 - Society's public records repository
 - A high-resolution tracking system for acknowledging human activity
 - A revolutionary organizing paradigm for human collaboration
 - An anticensorship mechanism
 - A liberty & equality enhancement tool.
 - A new organizing model for discovery
 - A " " " " for currency transfer
 - A coordination of all quanta or discrete units of anything
- Comprehending bcr at more generalized level with so many meanings conceptually helps to demonstrate its significant potential impact.

Currency, Token, Tokenizing

- Currency is just one idea that cryptoeconomy is forcing us to rethink.
- Traditional dictionary definition of currency is "A system of money in general use in a particular country".
- A secondary definition is more useful: "The quality or state of being used or accepted by many people".

This is more applicable for cryptocurrencies.

- What "backer" currency is high adoption rate, being accepted by many people. If more people were to accept the notion of cryptocurrencies & begin to use / trust them, they too could become liquid or fiat currencies.
- In cryptoeconomy context, one relevant way that word currency is used is in a generalized sense to connote "a unit of value that can be earned & used in certain economic system".

TOKEN;

- The nomenclature coin could just as easily be token - i.e.; a digital token or access or tracking mechanism for different activities.
- There could be Appcoin, communitycoin, AppToken, or other terms referring to different kinds of economic operations taking place within a community.

Counterparty Currency (xcp):

- It grants access to special features such as the ability to issue new assets, like new appcoin, etc.
- Counterparty protocol (xcp) is an economic system that facilitates anytime currency conversion to xcp or bitcoin, which is convertible to USD, EUR, CNY or any fiat currency.

xcp has 3 major components:

- The protocol: It is the tech that allows counterparty nodes to communicate with each other via bitcoin bc.
- The assets - It facilitates to create one's own digital tokens in a matter of seconds, which can represent anything and are easily exchanged peer-to-peer.
- xcp - The native currency of counterparty, used for advanced features & representing stake in protocol.

→ LTBCoin is a counterparty-enabled coin issued by Let's Talk Bitcoin media N/w to support its "local" economy.

- LTBCoin is used to transact incoming sponsorships, donations & tips & compensate outgoing content rewards, community participation acc, content creation, reviews & other forms of contribution.
- LTBCoin functions in the context of its own local economy, & is always immediately convertible to Bitcoin.
- Other currencies have similar use in their own local economies in the sense of interest community.
- One benefit of cryptocurrencies is their potential use as a tool for managing globally dispersed interest groups.

- Communitycoin like Boulder Farmers Market coin could provide additional features in its locality beyond just economic transactions, helping to build community cohesion & a more coordinated effort toward shared goals.
- Community cryptocoin could be a mechanism for increasing resolution of interest group activities by being a more specific means of organizing & coordinating group behavior toward some goal.

Community coin: Hayek's private currencies vie for attention:

- The explosion of altcoin & communitycoin, tokens or coins enabling economic function in a specific community context like LTBcoin.
- Denationalization of Money is a book by Austrian school economist Friedrich Hayek.
- He advocates a competitive private market for money instead of an arbitrary govt monopoly.
- He articulates other foundational thinking for bc industry by arguing against Keynesian inflationary money.
- Regarding decentralized currencies, he posits a model in which financial institutions issue their own currency & compete to maintain value of their currencies through productive activity.
- There can be multiple concurrent currencies. This model could be deployed on much wider basis in bc economy.

with the possibility that not just every financial institution, but every person, org & society would issue their own currency or token.

- The idea is that everyone could have their own coin, or multiple coins. Just like everyone has their own blog, Twitter & Instagram etc.
- Just like as everyone became an author in inform revolution everyone can become their own banker in bc revolution.

Campuscoin:

- Business & university campuses have their own economies for which currency issuance makes sense.
- There should be an open source, templated soft^m for any university to easily issue campuscoin.
- The templated altcoin could extend to groups within these communities.
- The campuscoin issuance template could have specific prepackaged modules.
 1. There could be a module for buying & selling assets within local community, an openBazaar or Craigslist like asset exchange module.
 2. There could be a sharing economy module, a decentralized model of Airbnb for dorm rooms, Getaround for transportation including cars & bikes & Lazooz peer-based ride sharing.
 3. There could be a consulting or "advisory services" module for all manner of advice, mentoring, coaching & tutoring related to classes, dept's, majors & careers.
 - Recent graduates could earn campuscoin by consulting

to job-seeking seniors with specific services like advice & mock interviews.

- campuscoin could provide a remunerative mechanism for these activities.

- It can be used to connect communities.

4. There could be a "peer-to-peer learning N/w" for notes sharing, book sharing, finding team members, forming study groups, studying for tests & providing other kinds of support.

5. There could be a RealJobs module connecting local employers with students for topical internships & jobs with industry exposure & job force readiness training, all in a reward-structured env.

There are several efforts under way to support students learning abt & using cryptocurrencies on university campuses.

Coin Drops as a Strategy for Public Adoption:

- The MIT Bitcoin project is effectively a coin drop, the simultaneous distribution of Bitcoin to entire populations to spur mainstream learning, trust & adoption.
- A larger-scale coin drop, BitDrop is scheduled for Caribbean island nation of Dominica for March 14, 2015 as part of Pi Day mathematical festival.
- Bitcoin will be sent by SMS via coinapult to all 70,000 residents.
- The goal is to create the world's largest & highest

demerit Bitcoin community.

- The project began as a brainstorming exercise to facilitate adoption & put Bitcoin into hands of as many as possible.
- Dominica was chosen as optimal b'coz the country has a relatively small population, a high cellular telephony penetration rate, & a position as a regional education center, and it is the center of an active intra-land, intra-currency trade & remittance economy.
- Coin drops or airdrops have been used in other situations.
- "Nationcoin" has been used to shore up national identity.
- Other coins are slotcoin, Spaincoin & Greececoin.
- Ecuador banned Bitcoin b'coz it plans to launch its own national cryptocurrency.
- Nationcoin could help bolster a sentiment of national patrimony, especially as many Eurozone nations have suffered from European central bank regulation impositions as a result of participating in Euro.
- The same kind of nationcoin benefits could be available in the idea of Tribecoin as patrimony-supporting coin issuance of native peoples.
- The Pine Ridge Indian Reservation in South Dakota was first American Indian tribe to launch its own cryptocurrency.
- Mazacoin, using tribal nation's sovereignty to set its own rules on cryptocurrencies.

Currency: New Meanings:

- The key point is that the term currency could begin to mean different things in cryptoeconomy context
 - More than in the basic money sense of serving as a payment mechanism for goods & services.
 - An imp sense of word currency in cryptoeconomy context is emerging as
 - "something of value that can be usefully deployed in some situation"
 - A unit of value that can be earned & used in a certain economic system".
 - There is general idea of token, currency or appcoin allowing access to certain features of an economic system.
 - Having Bitcoin for ex. allows access to performing transactions on BC.
 - Privileges are accorded to users in some cases just by holding Bitcoin, as this confirms ownership, and in other cases by ~~holding Bitcoin~~ spending Bitcoin.
 - Considering currency more broadly in these ways starts to widen its applicability to many other situations.
 - A currency is a token of value that can be earned & deployed.
 - It stores value & is transmissible.
 - This generalized definition supports that there can be many nonmonetary currencies that are conceived in same structure.
 - Reputation is a unit of value that can be earned & deployed in certain situations.

- Nonmonetary currency in the sense that it is a proxy for status or some kinds of tasks that a person can do.

Currency Multiplicity : Monetary & Nonmonetary currencies:

- multicurrency society with all kinds of monetary & non monetary currencies.
- First, there is currency multiplicity in the sense of monetary currency in that there are many diff fiat currencies.
- Second, there are many nonfiat, non-bc-based currencies like loyalty points & airline miles.
- There is multiplicity of bc-based cryptocurrencies like Bitcoin, Litecoin & Dogecoin.
- There is currency multiplicity in nonmonetary currencies. Such as reputation, intention & attention.
- Market Principles have been employed to develop metrics for measuring nonmonetary currencies such as influence, reach, awareness, authenticity, engagement, action taking, impact, spread, connectedness, velocity, participation, shared values & presence.
- BCT could make these nonmonetary social currencies more trackable, transmissible, transactable & monetizable.
- Social NW's could become social economic NW's.

Demurrage Currencies: potentially Incentive & Redistributable

- Currency is core concept in IoT that is being stretched, extended & reunderstood: currency as a digital token, a facilitation mechanism for quantized transfer.
- Demurrage means carrying cost - the cost to carry an asset.
- The term originated in freight & shipping industry to indicate the extra charge or cost associated with the detention in port of vessel by ship owner, in loading or unloading, beyond the time allowed or agreed.
- In cryptocurrency sense, demurrage can mean being deflationary over time, thus incentive in that it incites some form of action taking in shorter term to realize value before it is lost.
- Demurrage, is the compact concept of an attribute, the idea of an automatic motivating or incentive properly being built in to something.
- Another aspect of demurrage currencies is the notion of periodic automatic redistribution of currency across all v/w nodes at certain prespecified times or in certain events.
- Demurrage features could become a powerful & standard currency administration tool.
- Freicoin & Healthcoin are examples of uses of a demurrage currency with a built-in mechanism for action taking in form of spending.
- Demurrage currencies might be ideal for the implementation of guaranteed Basic Income initiatives (ABIs), systems

whereby all citizens or residents of a country would regularly receive an allowance - money sufficient to meet basic living expenses.

- A Bitcoin or Freecoin could be a straightforward currency for basic living expenses that runs out or resets on a periodic basis such as weekly, monthly or annually to keep system streamlined & efficient
- It is like a coupon, expiring after certain amount of time.
- A Bitcoin would not only currency, but would be a special use currency & would exist in context of multicurrency society.
- Different currencies for different purposes.